Analyzing Carbon Emissions Impacts on Global Temperature Using a Data-Driven Approach

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Youtube Link : https://youtu.be/HlSNsfAyk38

Abstract **- Climate change, driven by increasing carbon emissions, poses significant environmental, social, and economic challenges worldwide. This study, titled "Analyzing Carbon Emissions Impacts on Global Temperature Using a Data-Driven Approach," investigates the relationship between carbon emissions and global temperature anomalies to support evidence-based climate action. Using a comprehensive dataset spanning 1980–2021, which includes variables such as total CO_2 emissions, source-specific emissions (e.g., coal, oil, gas), energy production, and temperature anomalies across multiple regions, this project employs statistical and machine learning techniques to derive actionable insights. Pearson's correlation coefficient is utilized to quantify the strength and direction of the relationship between emissions and temperature anomalies, revealing significant correlations across regions and emission sources. Furthermore, Extreme Gradient Boosting (XGBoost) is leveraged to develop predictive models for future temperature anomalies, accounting for historical emission trends and regional factors. The findings highlight the disproportionate impact of fossil fuel-related emissions, particularly coal, on global warming, while also uncovering regional variations in climate response to emissions. This report provides policymakers and stakeholders with critical insights to inform climate mitigation strategies aligned with Sustainable Development Goal (SDG) 13: Climate Action. The research underscores the urgency of transitioning toward sustainable energy systems, reducing emissions, and implementing targeted interventions to mitigate the adverse effects of climate change.**

I. INTRODUCTION

Climate change poses a significant threat to global ecosystems, economies, and human health, primarily driven by anthropogenic activities such as industrialization, deforestation, and fossil fuel combustion. These activities contribute to the increasing concentration of greenhouse gases in the atmosphere, with carbon dioxide (CO_2) being the most dominant, accounting for about 75% of emissions [1]. As a result, the Earth's temperature has risen by over 1°C since pre-industrial times, leading to severe environmental consequences, including extreme weather events, rising sea levels, and biodiversity loss[2,3].

The Sustainable Development Goal (SDG) 13, Climate Action, underscores the importance of addressing these challenges through urgent measures to mitigate climate change and its impacts. Businesses, policymakers, and researchers are increasingly relying on data-driven approaches to analyze the relationship between carbon emissions and temperature anomalies, aiming to develop actionable strategies that align with the Paris Agreement's goal of limiting global temperature rise to 1.5°C–2°C[4].

This research focuses on the business domain of climate policy and sustainability, addressing how data analytics can inform decision-making processes. By examining historical trends in carbon emissions and their correlation with temperature anomalies across regions, this study aims to provide descriptive insights into "what is happening," predictive models for "what is likely to happen," and prescriptive strategies for "what should be done" to mitigate climate impacts effectively.

Key applications of this research include:

- Developing forecasting models to predict future temperature anomalies based on emission trends
- Providing insights for policy recommendations and sustainable practices in industries heavily contributing to carbon emissions
- Supporting the development of regional and global strategies for carbon reduction and climate adaptation.

By leveraging the comprehensive Global Energy and Emissions Dataset (1980-2021) from Kaggle, this study contributes to a better understanding of regional variations in climate response, fostering more targeted and effective climate action strategies.

II. BACKGROUND

Understanding the concepts used in this project is essential to analyzing the relationship between carbon emissions and global temperature anomalies effectively.

1. Pearson's Correlation Coefficient

One of the primary goals of this project is to explore the relationship between carbon emissions and temperature anomalies across regions and over time. Pearson's correlation coefficient, developed by Karl Pearson in the late 19th century, is a statistical measure widely used to assess the strength and direction of a linear relationship between two continuous variables [5].

This coefficient, denoted as *r*, ranges from -1 to 1. A value closer to 1 indicates a strong positive correlation, while values near -1 suggest a strong negative correlation. A value near 0 implies little to no linear relationship between the variables. Pearson's *r* also serves as a tool for hypothesis testing, allowing researchers to infer whether a significant relationship exists [6].

2. Extreme Gradient Boosting (XGBoost)

As part of this project, we aim to predict future temperature anomalies based on patterns observed in historical emissions data. Extreme Gradient Boosting (XGBoost), a machine learning algorithm developed by Chen and Guestrin in 2016, is particularly suited for this task due to its efficiency, accuracy, and scalability [7, 8].

XGBoost works by building an ensemble of decision trees where each tree corrects the errors of its predecessors. This iterative process minimizes a loss function, optimizing the model's predictions over time [9]. Additionally, the algorithm's ability to handle large datasets with high complexity makes it an ideal choice for this data-driven approach.

III. PROBLEM STATEMENT

Human-induced carbon emissions have emerged as one of the leading environmental challenges, driving a rise in global temperatures and exacerbating climate instability. Carbon dioxide levels have increased steadily over the past decades, reaching a record-high concentration of 419.3 ppm in 2023, which is over 50% higher than pre-industrial levels[3]. These emissions are linked to profound consequences, including higher rates

of extreme weather events, reduced agricultural productivity, and health-related vulnerabilities, particularly in low-income and middle-income countries [10].

While significant strides have been made in climate research, understanding the regional and sectoral impacts of carbon emissions on temperature anomalies remains a critical challenge. Variations in geographic sensitivity, industrial activity, and population density affect how emissions influence climate change across different regions. Moreover, the disproportionate effects of climate change on vulnerable populations emphasize the need for tailored mitigation strategies.

This study seeks to address the gap in understanding by investigating:

1. How carbon emissions influence temperature anomalies across regions.
2. The relative contribution of specific sources, such as fossil fuels, to temperature changes.
3. Regional disparities in climate response to emissions, providing insights for targeted interventions.

The findings aim to inform policymakers, businesses, and stakeholders in developing data-driven solutions for achieving SDG 13 goals, mitigating the adverse effects of climate change, and promoting sustainable practices globally.

IV. RESEARCH QUESTION/HYPOTHESIS

The research questions for the project includes:

1. What is the relationship between carbon emissions and global temperature anomalies across different regions over time?
2. Which sources of carbon emissions (e.g., coal, oil, gas) have the strongest correlations with temperature anomalies?
3. How do variations in population and industrial activity influence the relationship between carbon emissions and temperature anomalies across regions?
4. To what extent do fossil fuel-related emissions (coal, oil, and natural gas) contribute to global temperature increases compared to other emission sources?

The corresponding hypothesis for the questions are:

1. Increased carbon emissions are positively correlated with higher global temperature anomalies across regions and time, as CO_2 traps infrared heat and amplifies the greenhouse effect [11, 12].
2. Fossil fuel-related emissions (coal, oil, and gas) show the strongest correlations with temperature anomalies, with coal being the largest contributor due to its higher carbon content, followed by oil and gas [13, 14].
3. Regions with higher population density and greater industrial activity are likely to exhibit stronger relationships between carbon emissions and temperature anomalies, as these factors drive increased energy production and fossil fuel consumption.
4. Emissions from fossil fuel combustion have a disproportionately larger impact on temperature anomalies compared to other sources, as fossil fuels account for over 75% of global greenhouse gas emissions and nearly 90% of all CO_2 emissions [13, 14].

V. RESEARCH OBJECTIVES

1. To analyze the correlation between carbon emissions and global temperature anomalies across different regions over time.

2. To assess which sources of carbon emissions (e.g., coal, oil, gas) exhibit a stronger correlation with temperature anomalies.
3. To evaluate how population density and industrial activity, inferred through energy production and population data, influence the relationship between carbon emissions and temperature anomalies across regions.
4. To evaluate the relative influence of fossil fuel-related emissions (coal, oil, and gas) compared to other emission sources on temperature changes.

VI. RESEARCH SIGNIFICANCES

1. Understanding how carbon emissions influence global temperature anomalies across regions provides policymakers with actionable data to create region-specific and globally aligned strategies.
2. Insights into the sources of carbon emissions, such as fossil fuels, enable industries to identify high-impact areas for emission reduction. This encourages investment in greener technologies and promotes sustainable industrial practices.
3. Developing forecasting models for temperature anomalies empowers governments, businesses, and communities to anticipate and prepare for climate-related challenges, thereby reducing vulnerability and enhancing resilience.
4. Identifying disparities in regional climate responses to emissions allows for the development of targeted mitigation strategies, ensuring equitable and efficient resource allocation.

VII. LITERATURE REVIEW

No	Year	Authors	Research Problem	Main Techniques	Results	Future Works
1	2019	Nunez, NOAA	Analysis of carbon emissions trends and their impact on global temperatures	Statistical analysis, temperature trend modeling	Identified a strong positive correlation between increasing CO2 levels and global temperature rise	Extend the analysis to include other greenhouse gases like methane and nitrous oxide
2	2019	Lancet Countdown	Assessing the health and environmental effects of climate change	Health impact modeling, policy review	Highlighted the urgent need for region-specific climate adaptation strategies	Explore the long-term health impacts of delayed climate actions
3	2024	Fecht, Nat. Hist.	Explanation of greenhouse effect mechanisms	Simulation and theoretical modeling	Demonstrated how CO2 enhances the greenhouse effect, intensifying global warming	Investigate the role of cloud cover and aerosols in modulating the greenhouse effect
4	2023	Client Earth, UN	Impact of fossil fuels on carbon emissions	Emission source analysis, case studies	Found that coal, oil, and gas are the dominant contributors to CO2 emissions	Propose alternative energy solutions and evaluate their feasibility in high-emission countries
5	2022	Various (Regional Studies)	Regional analysis of emission-related temperature anomalies	Geospatial data analysis, regression	Revealed significant regional disparities in temperature responses to emissions	Expand the study to include socioeconomic factors affecting regional climate vulnerability
6	2019	Alexander, MacIsaac, Matthe	The greenhouse gas climate commitment and	Climate modeling,	Highlighted the irreversible nature	Explore mitigation strategies to

		ws, Mengis, Zickfeld	reversibility of peak warming from historical emissions	historical emission analysis	of certain warming due to past emissions, even with immediate emission cuts	address committed warming and assess the potential for carbon removal technologies
7	2024	Filonchyk, Peterson, Zhang, Hurynovich, He	Greenhouse gases emissions and global climate change: Examining the influence of CO2, CH4, and N2O	Statistical analysis, comparative gas impact study	Found that CH4 and N2O have higher short-term warming potential, though CO2 remains the largest long-term contributor	Investigate mitigation techniques specific to CH4 and N2O while optimizing CO2 reduction strategies
8	2022	Zilin, Su	World CO2 Emissions: Simple Analysis and its Relationship with Global Temperature Change	Regression analysis, temperature-emission modeling	Demonstrated a direct, positive relationship between CO2 emissions and global temperature rise over time	Examine the role of non-CO2 factors like aerosols and land use in modulating temperature changes
9	2024	Ivanchenko, A. V.	Shifting the focus of world industrial development from European countries and North America to Asia	Geospatial analysis, industrial activity review	Identified Asia as the new hub of industrial emissions, with notable reductions in European and North American contributions	Evaluate the environmental impact of shifting industrial activities and propose sustainable industrialization practices
10	2023	Ayodele, Idowu, Obaika, Ohikhuare, Ahmad, Chowdhury	Does industrialization trigger carbon emissions through energy consumption? Evidence from OPEC and high-industrialized countries	Econometric modeling, energy consumption analysis	Found a strong link between industrialization and carbon emissions, mediated by energy consumption patterns	Develop policies to decouple energy-intensive industrialization from emissions by promoting renewable energy adoption

VIII. METHODOLOGY

Dataset Description

The dataset was obtained from Kaggle titled ‘Global Energy and Emissions Dataset (1980–2021)’. The dataset includes two parts (tables), first country-level data with annual statistics for 211 countries and second global aggregates offering summary metrics for worldwide trends. This dataset shows changes over time in CO_2 emissions, energy use, methane emissions, population growth, and temperature anomalies at both global and country levels. It combines data from reputable sources like CERN, EIA, EDGAR, the World Bank, and IMF, providing a comprehensive view of human activity and its environmental impact. As the data was collected worldwide, it allowed us to understand past trends and their possible connections. Providing insights into climate change, policy impacts, and the links between human activity (gas emissions), energy use, population growth and temperature anomalies.

Tools Used

The instrument employed for analysing this dataset was Google Colab, an online integrated development environment (IDE) that offers a Python-based platform for data analysis and machine learning. Google Colab is well designed for collaborative and interactive data exploration, providing seamless access to robust computing resources, including GPUs and TPUs, without the need for local

configuration. Python was employed in Google Colab due to its adaptability and comprehensive libraries such as pandas for data management, matplotlib and seaborn for visualisation, and numpy for numerical analysis. This combination of technologies facilitated effective exploration, visualisation, and analysis of the dataset, rendering it an optimal selection for this research.

Algorithms Implemented

1. Correlation Analysis

   Statistical approaches like Pearson correlation were utilised to measure the linear correlations between variables such as carbon emissions, temperature anomalies, and population. This helped identify substantial links between emission sources and their impact on global temperature fluctuations.

2. Regression Analysis

   Multiple regression models were utilised to analyse how different emission sources (coal, oil, and gas) and characteristics like population affect temperature anomalies. This included linear regression to anticipate the impact of emissions and interaction terms to assess complex interactions between factors.

3. XGBoost Regression

   A powerful machine learning technique, XGBoost (Extreme Gradient Boosting), was used to predict temperature anomalies based on the feature total CO_2 emissions. XGBoost is renowned for its speed, scalability, and capacity to handle big datasets while capturing non-linear correlations.

4. Classification Model

   Temperature changes were sorted into levels (e.g., low, moderate, high). And a XGB classifier was used to predict the temperature category based on the total carbon emission. Techniques like label encoding were employed to prepare data, followed by classification to predict the temperature change category.

Training and Testing Dataset

The project splits the dataset into training and testing subsets to evaluate the performance of its machine learning models. The training dataset is used to train algorithms like XGBoost for forecasting temperature anomalies based on the parameter total CO_2 emissions. The testing dataset acts as an unseen set of data to assess the model's accuracy and generalization capabilities. The split was accomplished using a conventional ratio of 80% for training and 20% for testing, ensuring that both groups match the data distribution well.

Performance Measurements

The XGBoost regression and classification models were evaluated using the following metrics:

Classification:

1. Accuracy

   Represents the proportion of correctly classified instances compared to the total number of instances evaluated in the classification task for categorizing temperature changes (e.g., low, moderate, high).

$$Accuracy = \frac{TP + TN}{Total\ Instance}$$

2. Precision, Recall, and F1-Score

   Precision measures the model's ability to reduce false positives, while recall assesses its effectiveness in identifying all positive instances, making these metrics crucial for evaluating performance in identifying specific categories, particularly when certain temperature ranges are less frequent. The F1 Score balances precision and recall, providing a single metric that reflects the model's ability to avoid false positives and identify true positives, especially in imbalanced datasets.

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Where, TP (True Positive) are correctly predicted positives, FP (False Positive) are incorrectly predicted positives, FN (False Negative) are missed positives, and TN (True Negative) are correctly identified negatives.

Regression:

1. Mean Squared Error (MSE)

   Measures the average squared difference between the predicted and actual temperature anomalies, penalizing larger errors more heavily.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

2. Mean Absolute Error

   Captures the average magnitude of prediction errors, providing a straightforward interpretation of model accuracy.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \hat{y}_i\right|$$

3. R-squared (R^2)

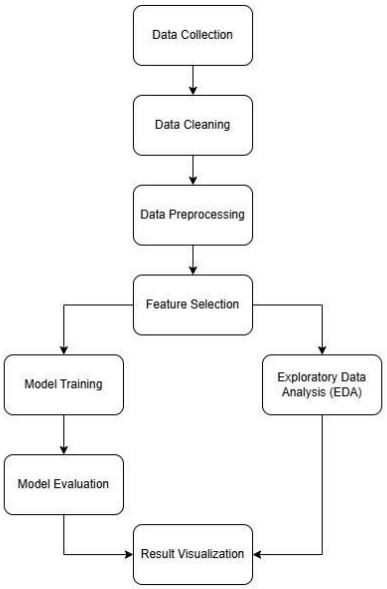
   Indicates the proportion of variance in the target variable (temperature anomalies) explained by the features, reflecting the model's goodness of fit.

$$R^2 = \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

- $y_i = Actual\ value\ of\ the\ target\ variab$
- $\hat{y}_i = Predicted\ value\ by\ the\ model$
- $\overline{y_i} = Mean\ of\ the\ actual\ values$
- N = Number of observations

Model Diagram

The flow model is summarized in the flow diagram below:



Data Cleaning and Preprocessing

Data cleaning and preprocessing involves preparing data for analysis by handling missing values, correcting errors, and transforming data into a suitable format. This process ensures the quality and consistency of the dataset, making it ready for modeling and analysis.

Handling Missing Values:

Initially, we checked for missing values in both global and country tables using the commands as shown below. And found that there were no missing values.

```
# Check for missing values
print("\nMissing Values in Global Dataset:")
print(global_data.isnull().sum())

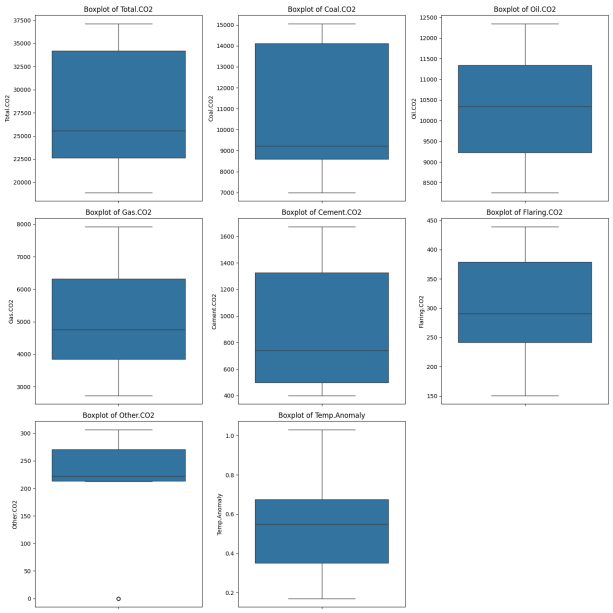
print("\nMissing Values in Country Dataset:")
print(country_data.isnull().sum())
```

```
Missing Values in Global Dataset:
Year                                  0
Country                               0
ISO.alpha-3                           0
Total.CO2                             0
Coal.CO2                              0
Oil.CO2                               0
Gas.CO2                               0
Cement.CO2                            0
Flaring.CO2                           0
Other.CO2                             0
Per.Capita.CO2                        0
Total.Energy.Production               0
Coal.Energy                           0
Gas.Energy                            0
Petroleum.and.other.liquids.Energy    0
Nuclear.Energy                        0
Renewables.and.other.Energy           0
Temp.Anomaly                          0
dtype: int64

Missing Values in Country Dataset:
Unnamed: 0                            0
Year                                  0
Country                               0
ISO.alpha-3                           0
Total.CO2                             0
Coal.CO2                              0
Oil.CO2                               0
Gas.CO2                               0
Cement.CO2                            0
Flaring.CO2                           0
Other.CO2                             0
Per.Capita.CO2                        0
Total.Energy.Production               0
Coal.Energy                           0
Gas.Energy                            0
Petroleum.and.other.liquids.Energy    0
Nuclear.Energy                        0
Renewables.and.other.Energy           0
CH4                                   0
Population                            0
Temp_Change                           0
dtype: int64
```

Outlier Detection and Handling Duplicate:

Outlier detection was performed using the IQR method and box whisker plots to identify and mitigate the impact of extreme values as shown below.




```
Column: Total.CO2
Number of outliers: 0
Lower bound: 5251.451183750003, Upper bound: 51579.28182974999

Column: Coal.CO2
Number of outliers: 0
Lower bound: 349.54950587499843, Upper bound: 22370.395476875

Column: Oil.CO2
Number of outliers: 0
Lower bound: 6058.251025250002, Upper bound: 14510.374787249999

Column: Gas.CO2
Number of outliers: 0
Lower bound: 144.6545444999988, Upper bound: 10031.627418500002

Column: Cement.CO2
Number of outliers: 0
Lower bound: -745.7827196250003, Upper bound: 2569.1802773750005

Column: Flaring.CO2
Number of outliers: 0
Lower bound: 34.286479749999984, Upper bound: 586.06448375

Column: Other.CO2
Number of outliers: 10
Lower bound: 127.84250537499997, Upper bound: 356.921334375
```

Duplicate rows also were checked and removed to prevent redundancy as shown below.

```
# Check for duplicate rows in the Global Dataset
global_duplicates = global_data.duplicated().sum()
print(f"Number of duplicate rows in Global Dataset: {global_duplicates}")

# Check for duplicate rows in the Country Dataset
country_duplicates = country_data.duplicated().sum()
print(f"Number of duplicate rows in Country Dataset: {country_duplicates}")

# Remove duplicate rows if any exist
if global_duplicates > 0:
    global_data.drop_duplicates(inplace=True)
    print("Duplicates removed from Global Dataset.")

if country_duplicates > 0:
    country_data.drop_duplicates(inplace=True)
    print("Duplicates removed from Country Dataset.")
```

```
Number of duplicate rows in Global Dataset: 0
Number of duplicate rows in Country Dataset: 0
```

Feature Selection:

Feature selection involved dropping irrelevant or redundant columns (e.g., identifiers like Country codes) to focus on variables contributing to the analysis as shown below.

```
# Drop unwanted columns.
unwanted_columns_global = ['Country','ISO.alpha-3','Per.capita.CO2','Total.Energy.Production','Coal.Energy','Gas.Energy','Petroleum.and.other.liquids.Energy',
                           'Nuclear.Energy','Renewables.and.other.Energy','Population']
unwanted_columns_country = ['ISO.alpha-3','Total.Energy.Production','Coal.Energy','Gas.Energy','Petroleum.and.other.liquids.Energy',
                            'Nuclear.Energy','Renewables.and.other.Energy']

global_data.drop(columns=unwanted_columns_global, inplace=True, errors='ignore')
country_data.drop(columns=unwanted_columns_country, inplace=True, errors='ignore')
```

```
# Display the first few rows of the global dataset.
print("First few rows of the Global Dataset:")
global_data.head()
```

First few rows of the Global Dataset:

	Year	Total.CO2	Coal.CO2	Oil.CO2	Gas.CO2	Cement.CO2	Flaring.CO2	Other.CO2	Temp.Anomaly
0	1980	19501.263339	7024.128811	8951.775558	2758.101427	402.619608	316.435558	0.0	0.29
1	1981	19036.118109	7004.883319	8550.304967	2794.346721	402.748288	237.162981	0.0	0.35
2	1982	18884.487014	7126.512351	8345.175954	2731.273279	400.937145	235.574963	0.0	0.19
3	1983	19006.539624	7314.190173	8260.815187	2760.541893	409.876965	212.846811	0.0	0.35
4	1984	19664.753809	7668.797447	8328.840383	3010.425483	417.702984	186.751322	0.0	0.20

```
# Display the first few rows of the country dataset.
print("\nFirst few rows of the Country Dataset:")
country_data.head()
```

First few rows of the Country Dataset:

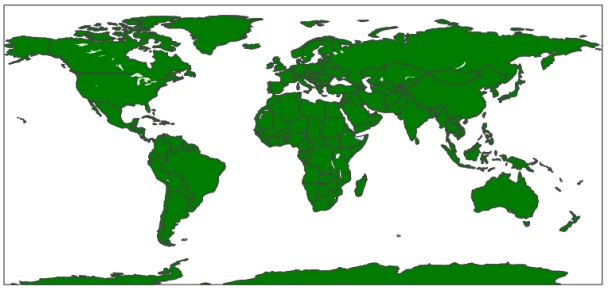
	Unnamed: 0	Year	Country	Total.CO2	Coal.CO2	Oil.CO2	Gas.CO2	Cement.CO2	Flaring.CO2	Other.CO2	Per.Capita.CO2	CH4	Population	Temp_Change
0	1	1980	Afghanistan	1.756302	0.315782	0.925256	0.187254	0.023284	0.304747	0.0	0.140655	456.9363	12486631	0.600
1	2	1981	Afghanistan	1.978463	0.333424	1.014828	0.304112	0.032879	0.293120	0.0	0.177358	452.3030	11155195	0.483
2	3	1982	Afghanistan	2.094581	0.384720	0.992944	0.395712	0.039077	0.282128	0.0	0.207625	446.6548	10088289	-0.346
3	4	1983	Afghanistan	2.519954	0.384720	1.220112	0.615552	0.006450	0.293120	0.0	0.253225	424.4818	9951449	0.164
4	5	1984	Afghanistan	2.821540	0.392556	1.133644	0.931863	0.047963	0.315513	0.0	0.275442	382.8322	10243686	0.145

Exploratory Data Analysis (EDA)

In order to gain a better understanding of the dataset before conducting a more in-depth analysis, we performed Exploratory Data Analysis (EDA) to expose general trends, correlations, and notable global insights. The EDA phase included the following steps:

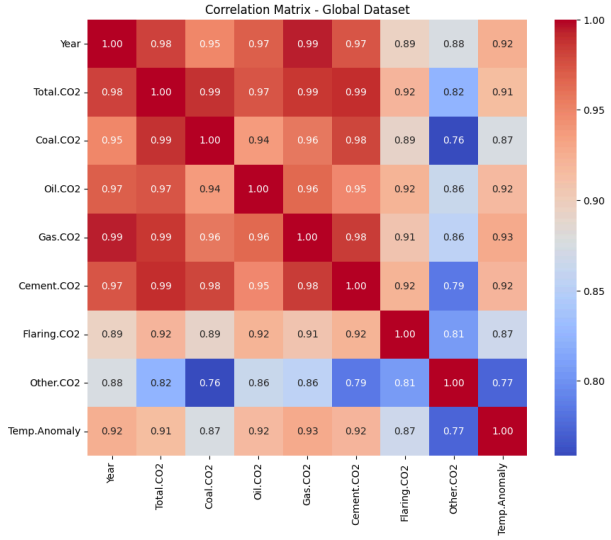
1. **Unique Countries:** We retrieved the number and list of unique countries in the dataset using nunique() and unique(), and created a geographical visualization with a choropleth

map in Plotly to highlight countries represented in the dataset as green.



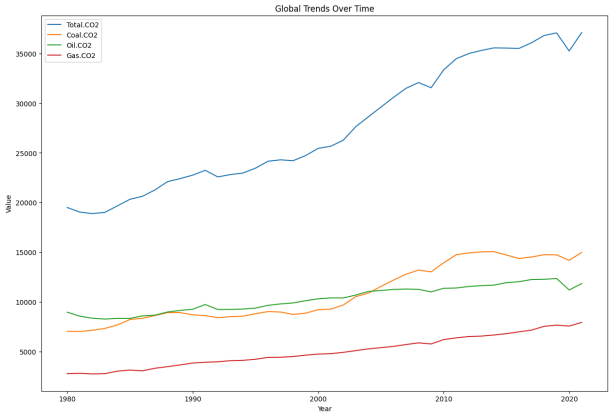
2. **Correlation Matrix:** We visualized the correlation matrix for the global table with a heatmap to examine relationships between numerical variables. This heat map showed that fossil fuel CO2 emissions tend to have a higher correlation with temperature anomaly.

```
[ ] # Correlation matrix for Global Dataset
    plt.figure(figsize=(10, 8))
    corr_matrix = global_data.corr()
    sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f")
    plt.title("Correlation Matrix - Global Dataset")
    plt.show()
```



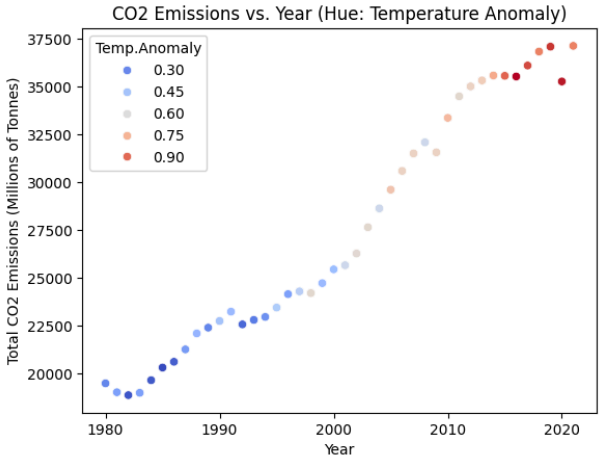
3. **Trends Over Time**: A time-series plot was created to examine trends in emissions and temperature changes over time. The plot showed that coal and oil had the highest C02 emissions rates.

```
[ ] # Plot trends over time for key variables
    time_columns = ['Total.CO2', 'Coal.CO2', 'Oil.CO2', 'Gas.CO2']
    plt.figure(figsize=(15, 10))
    for col in time_columns:
        sns.lineplot(data=global_data, x='Year', y=col, label=col)
    plt.title("Global Trends Over Time")
    plt.xlabel("Year")
    plt.ylabel("Value")
    plt.legend()
    plt.show()
```



4. **Scatter Plot for Relationships**: a scatter plot was used to investigate the relationship between total CO2 emissions and temperature anomaly over the years. It showed that they had a strong positive correlation.

```
[ ] # Scatter plot to observe relationships and potential outliers
    sns.scatterplot(data=global_data, x='Year', y='Total.CO2', hue='Temp.Anomaly', palette='coolwarm')
    plt.title("CO2 Emissions vs. Year (Hue: Temperature Anomaly)")
    plt.xlabel("Year")
    plt.ylabel("Total CO2 Emissions (Millions of Tonnes)")
    plt.show()
```



IX. RESULTS

This section presents the findings from the dataset in addressing our research questions, which are as follows:

1. What is the relationship between carbon emissions and global temperature anomalies across different regions over time?
2. Which sources of carbon emissions (e.g., coal, oil, gas) have the strongest correlations with temperature anomalies?
3. How do variations in population and industrial activity influence the relationship between carbon emissions and temperature anomalies across regions?
4. To what extent do fossil fuel-related emissions (coal, oil, and natural gas) contribute to global temperature increases compared to other emission sources?

Relationship Between carbon emissions and global temperature anomalies across different regions over time.

This part explains the first research question: What is the relationship between carbon emissions and global temperature anomalies across different regions over time? To address this, Figure 1 presents the trend of regional CO_2 emissions over time, while Figure 2 illustrates the corresponding regional temperature over time. By analyzing these figures, regional trends in CO_2 emissions are compared with temperature anomalies, highlighting the potential correlation between emissions and temperature changes. This comparison provides a clearer understanding of the dynamic relationship between emissions and global warming over time.

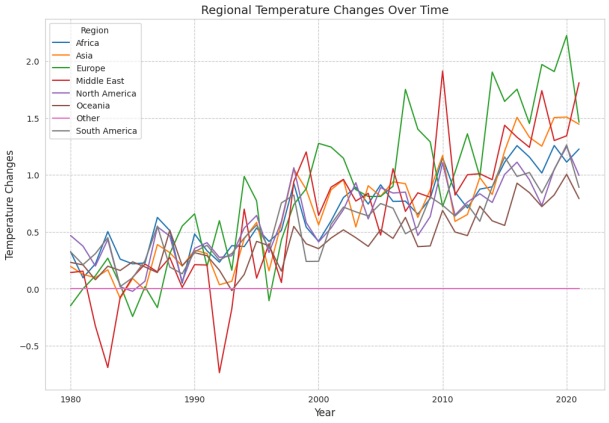


Figure 1. Regional Temperature Changes Over Time

Figure 1 illustrates regional temperature changes over time, results show 90% of the regions had temperature rises over time. This trend is potentially attributable to rising CO_2 emissions.

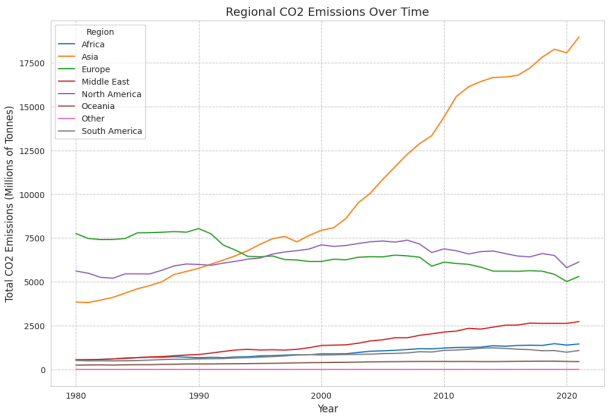


Figure 2. Regional CO2 Changes Over Time

Figure 2 illustrates regional CO_2 emissions over time,focusing on North America and Europe.The data shows North America has seen a slight increase in CO_2 emissions and Europe has experienced a small decrease, but still temperatures over time have continued to rise. This is due to the long atmospheric lifetime of CO_2, which remains in the atmosphere for centuries .This can be explained by historical emissions that have contributed significantly to current warming, with CO_2 remaining in the atmosphere for centuries, leading to a long-term climate commitment [15].

The relationship between carbon emissions and global temperature anomalies across different regions over time is largely because of the release of greenhouse gases, including CO_2, CH_4, and N_2O. These gases are primarily emitted through human activities such as fossil fuel combustion, agriculture, and industrial processes [16]. The industrial revolution marked a significant turning point, leading to a substantial increase in CO_2 levels, which disrupted the natural carbon cycle and contributed to the ongoing trend of global warming [17].

Correlations between sources of carbon emissions (e.g., coal, oil, gas)with temperature anomalies.

This part explains the research question 2 : Which sources of carbon emissions (e.g., coal, oil, gas) have the strongest correlations with temperature anomalies? By using a subplot of all the source CO2, we can find which sources have the strongest correlation with temperature anomalies.

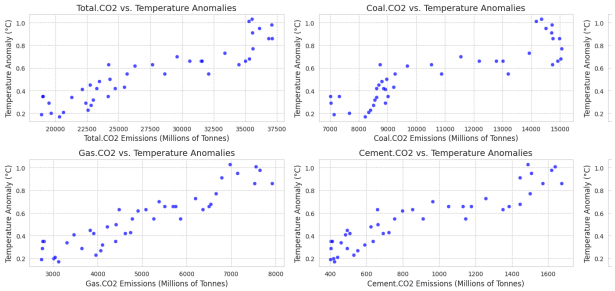


Figure 3. Sources of CO2 and Global Temperature

The subplots show the relationship between various sources of CO2 emissions and global temperature anomalies over time. Total CO2, coal, gas, and cement emissions demonstrate a strong correlation with temperature anomalies.CO2 emissions from coal clearly stand out as a primary contributor to the global temperature rise compared to emissions from gas and cement.

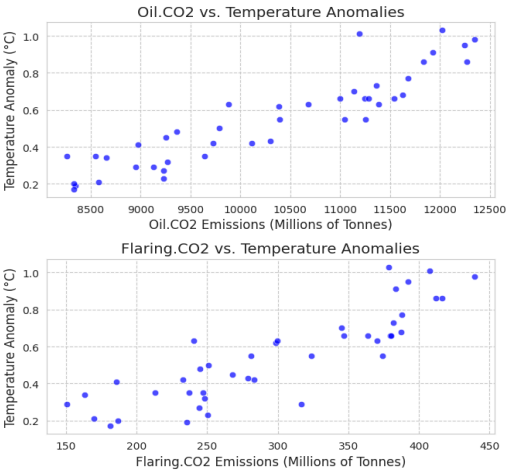


Figure 4. Sources of CO2 and Global Temperature

Figure 4 illustrates the relationship between Oil CO2 and Flaring CO2 emissions and temperature anomalies. Oil CO2 shows a significantly higher amount of CO2 emissions compared to Flaring CO2. This indicates that oil is also a primary contributor to temperature anomalies.

The analysis of CO_2 emissions from various sources reveals that both coal and oil play significant roles in contributing to global temperature anomalies. This is mainly because the amount of CO_2 produced from oil and coal is the highest among all other sources.

Population and Industrial Activity's Impact on Carbon Emissions and Temperature Anomalies.

This part explains the research question 3 :How do variations in population and industrial activity influence the relationship between carbon emissions and temperature anomalies across regions? By analysing two figures which are figure 5: population and CO2 emission and figure 6 : population and Temperature,we can identify the impact of population and industrial activity on carbon emissions and temperature anomalies.

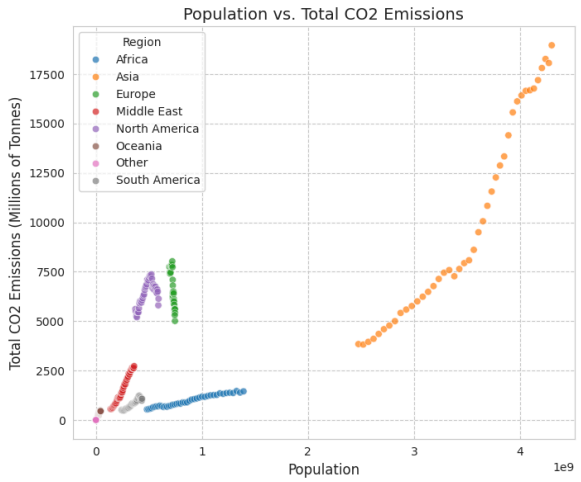


Figure 5. Population and total number of CO2

The graph illustrates various regional populations and the total number of CO2 emissions. We can analyse regions with higher populations, such as Asia, that exhibit significantly larger total CO2 emissions compared to less populated regions.The region like North america and europe despite having a smaller population but still have moderate number of CO2 emission.

The increasing slope of CO2 emissions is due to highly industrialized regions such as North America and Europe, suggesting that industrial activity significantly amplifies CO2 emissions [18] [19]. and for less industrialized regions (e.g., Africa), emissions remain relatively low despite rising population.

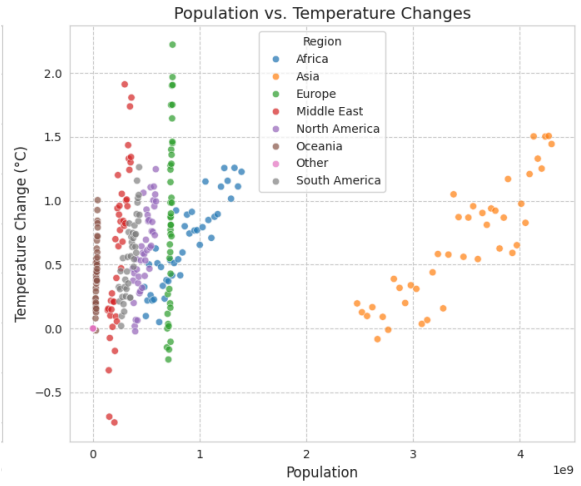


Figure 6. Population and temperature changes

Figure 6 shows the global population and temperature changes. Asia has a large population but does not exhibit significant differences in temperature changes compared to other regions. This is solely because temperature anomalies are not as strongly correlated with population as they are with CO2 emissions and are more likely influenced by cumulative historical emissions and localized environmental factors.

Impact of Fossil Fuels and Other Emissions on Global Temperature.

This part explains the last research question 4: To what extent do fossil fuel-related emissions (coal, oil, and natural gas) contribute to global temperature increases compared to other emission sources?

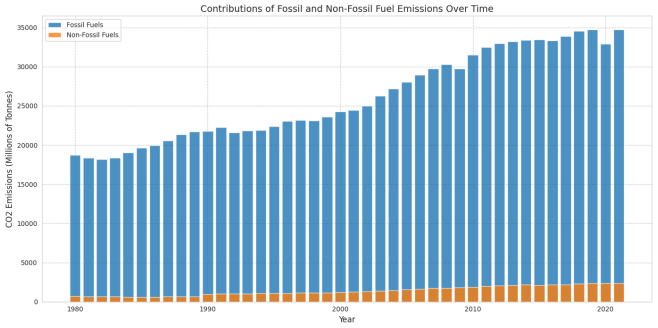


Figure 7. Contribution of Fossil and Non-Fossil Fuel to CO2 Emission Over Time

The graph illustrates the contribution of fossil fuels and non-fossil fuels to CO2 emissions over time. By analyzing the graph, it is clear that fossil fuels contribute the most to CO2 emissions over time compared to non-fossil fuels.

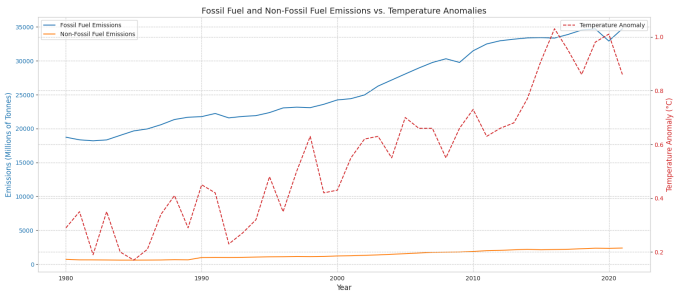


Figure 7. Fossil and Non-Fossil Fuel and Temperature Anomalies Over Time

The graph illustrates a clear upward trend in temperature anomalies corresponding to the increase in fossil fuel emissions. In contrast, non-fossil fuel emissions have remained minimal and show no significant correlation with global temperature increases. By analyzing this graph, we can see that fossil fuel-related emissions are the main factor contributing to the rise in global temperatures, compared to other emission sources.

Model Evaluation

1. Regression Results

   To predict temperature anomalies using total CO_2 emissions, the XGBoost regression model was employed. The performance of the regression model is summarized in Table 1. The relatively low R^2 score indicates that the model captures only a small portion of the variability in the data, suggesting opportunities for further feature engineering or model tuning.

Table 1: Regression Model Evaluation Metrics

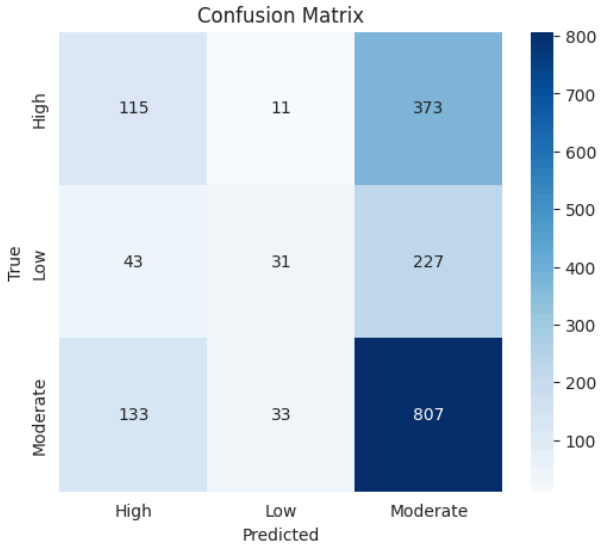
Metric	Value
Mean Squared Error	0.3833
Mean Absolute Error	0.4820
R^2 Score	0.0232

2. Classification Results

   To classify temperature changes into categories (low, moderate, high), an XGBoost classifier was utilized. The classification model achieved moderate performance, as shown in Table 2. The confusion matrix below highlights the distribution of true and predicted labels, indicating that the model performs best at predicting the "Moderate" category but struggles with "High" and "Low."

Table 2: Classification Evaluation Metrics

Metric	Value
Accuracy	0.5375
Precision	0.4962
Recall	0.5375
F1-Score	0.4800



X. DISCUSSIONS

The results of our analysis shed light on the intricate relationships between carbon emissions, global temperature anomalies, population growth, industrial activity, and the contributions of various emission sources. Below, we discuss each finding, contextualizing its implications for policymakers, industries, and global climate strategies.

Relationship Between Carbon Emissions and Global Temperature Anomalies Across Regions Over Time

Our findings reveal a significant association between rising carbon emissions and temperature anomalies across various regions. Figures 1 and 2 illustrate that 90% of the studied regions experienced temperature increases over time, with North America and Europe displaying persistent warming trends despite

slight reductions in their CO_2 emissions. This emphasizes the long atmospheric lifetime of CO_2 and its lasting impact on global temperatures.

The observed trends underscore the critical need for global coordination in reducing emissions, particularly focusing on historical emitters with substantial contributions to present-day warming. Policymakers could use these insights to advocate for long-term carbon reduction strategies, enforce stricter emission caps, and invest in carbon sequestration technologies to mitigate ongoing climate commitments.

Correlation Between Emission Sources and Temperature Anomalies

Our analysis identifies coal and oil as the most significant contributors to global temperature anomalies among the studied emission sources. Figure 3 highlights the strong correlation between coal-based emissions and temperature anomalies, while Figure 4 demonstrates that oil emissions rank just behind coal in terms of their impact. The relatively lower contributions of gas and cement emissions are evident in their comparatively weaker correlations.

These findings highlight the urgent need to transition away from coal and oil as dominant energy sources. Industries and governments can prioritize renewable energy development, impose higher taxes on coal and oil usage, and incentivize clean energy investments. Additionally, global climate agreements could target these emission sources for aggressive reduction to limit further warming.

Influence of Population and Industrial Activity on Carbon Emissions and Temperature Anomalies

The analysis of population and industrial activity reveals distinct patterns in their impact on emissions and temperature anomalies. Figures 5 and 6 show that regions with higher populations, such as Asia, exhibit significantly larger CO_2 emissions, yet temperature anomalies remain relatively uniform across regions. This suggests that temperature changes are more influenced by cumulative historical emissions than by population growth alone.

In contrast, highly industrialized regions like North America and Europe show substantial emissions relative to their smaller populations. This finding underscores the role of industrial activity in amplifying emissions and highlights the importance of adopting cleaner industrial practices. Policymakers in highly industrialized regions should focus on stricter environmental regulations, promote green technologies, and support sustainable industrial development to mitigate emissions without compromising economic growth.

Impact of Fossil Fuel-Related and Other Emissions on Global Temperature

The analysis of fossil fuel-related emissions, as indicated by the trends in coal, oil, and gas usage, reveals their dominant role in driving temperature anomalies. While alternative sources like flaring and cement emissions contribute less significantly, the cumulative effect of fossil fuels remains a primary driver of global warming. This underscores the need for urgent global action to reduce reliance on fossil fuels and transition to sustainable energy alternatives.

These insights emphasize the necessity of diversifying energy portfolios, investing in renewable energy infrastructure, and enacting stringent emission reduction policies. International collaboration, particularly in regions heavily reliant on fossil fuels, is critical to achieving long-term climate goals.

XI. FUTURE WORKS

Although our study shows the strong correlation between global temperature anomalies and carbon emissions, stressing the contributions of several

emission sources including coal, oil, and gas helps. Few future projects could be focused to improve the scope of analysis and the forecasting powers. Including more complete dataset, such as those documenting trends in renewable energy sources, rates of deforestation, or technical development. It may help one have a better knowledge of the elements causing variations in world temperature. Next, to improve prediction accuracy, future work could explore using advanced machine learning models, such as neural networks or methods that combine multiple models, like ensemble approaches. Other than that, geospatial methods could help to extend regional-level studies by capturing local effects of carbon emissions. Furthermore, including real-time data streams and creating interactive dashboards will let legislators and scientists dynamically track temperature variations and emissions. Therefore, increasing public knowledge of results by means of visualizing tools could help encourage more action on climate change.

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